



Karnataka State Police
Government of Karnataka

Powered by **HACK2SKILL**

Dataathon

2026

Technology partner



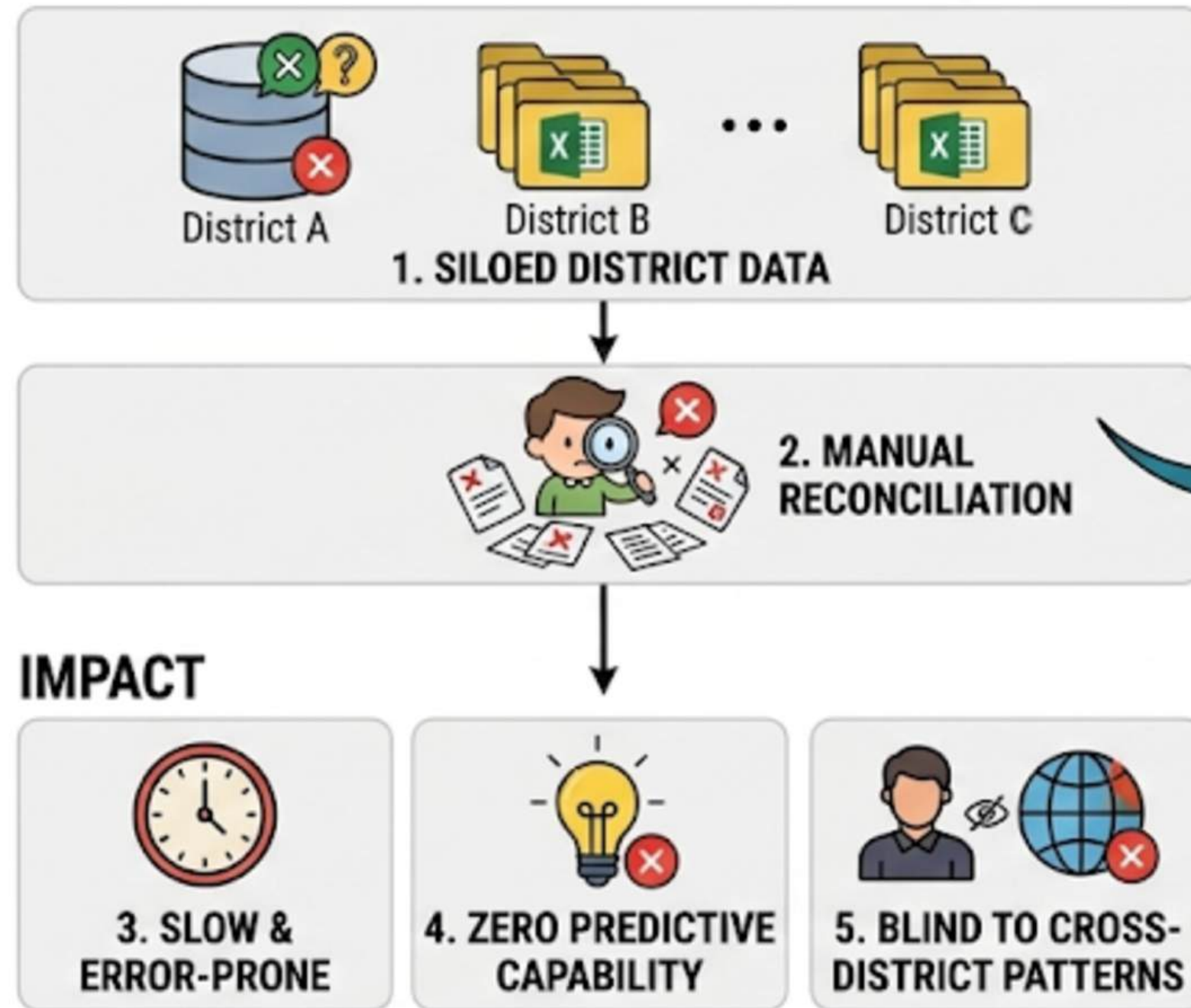
Team Details

- a. Team name: Coder4not4**
- b. Team leader name: Mukul Prasad**
- c. Team size: 3**
- d. Problem Statement: AI-Driven Crime Analytics & Visualization Platform**

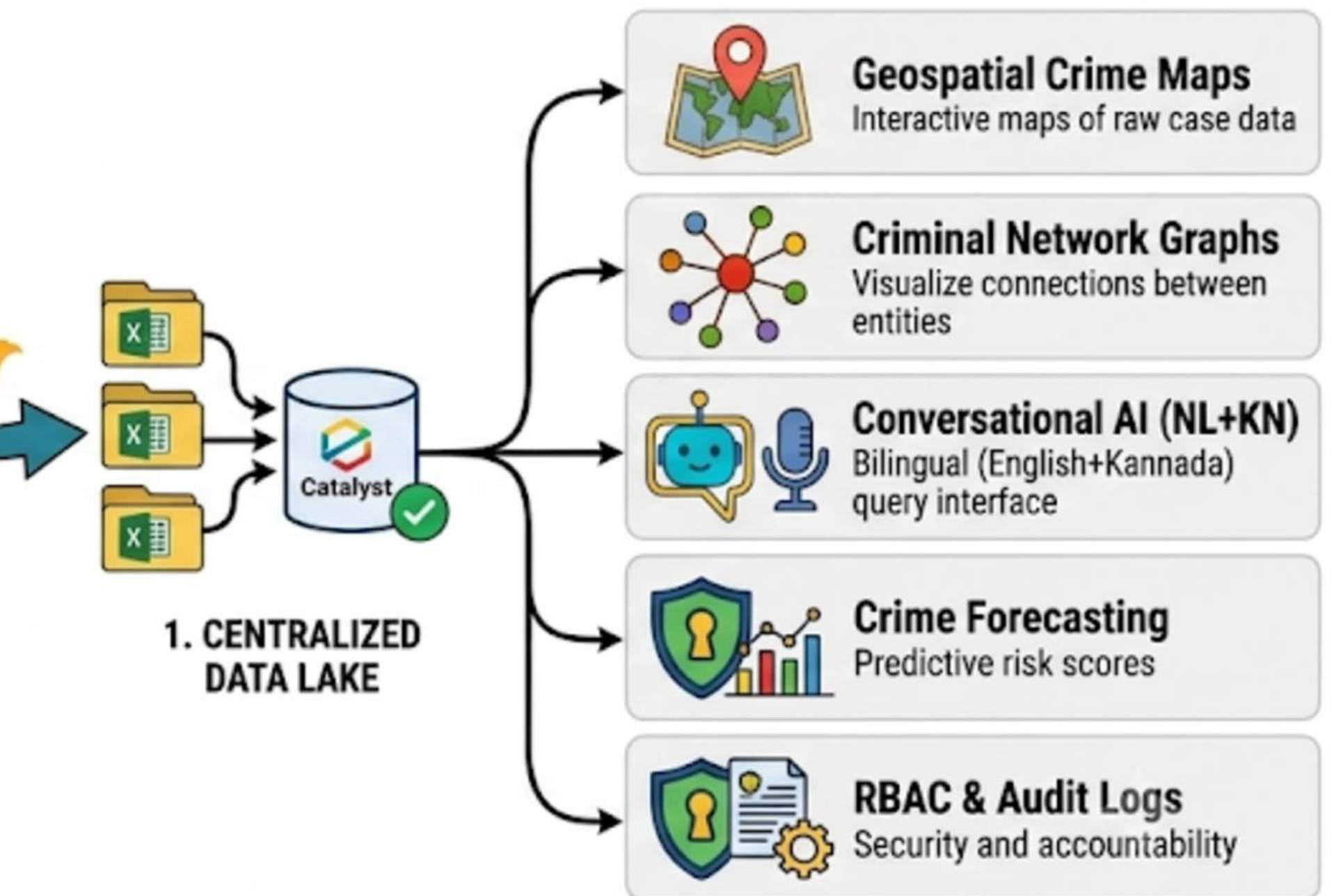


Brief about the solution

The Problem: Siloed Data & Manual Analysis



Our Solution: Integrated Intelligence Platform (Project Falcon)



No external AI APIs. All compute, ML, storage & orchestration runs inside Zoho Catalyst.

Opportunities

Dimension	Traditional Approach	Project Falcon
AI Processing	Third-party LLM APIs (OpenAI etc.)	100% Zoho Catalyst — zero external AI calls
Data Structure	Excel sheets, per-district silos	Centralised 28-table DataStore + Evidence Registry
Analytics Depth	Manual counts, static charts	DBSCAN / K-Means/ SARIMA / Louvain / TF-IDF
Policing Posture	Reactive — after incident	Proactive — 7/30-day forecasts + anomaly alerts
Explainability	None — no source trace	Every AI answer cites source FIR number(s)



Evidence Trail — Every AI-generated answer carries the source FIR citation number, making it legally traceable and suitable for investigative and judicial use.



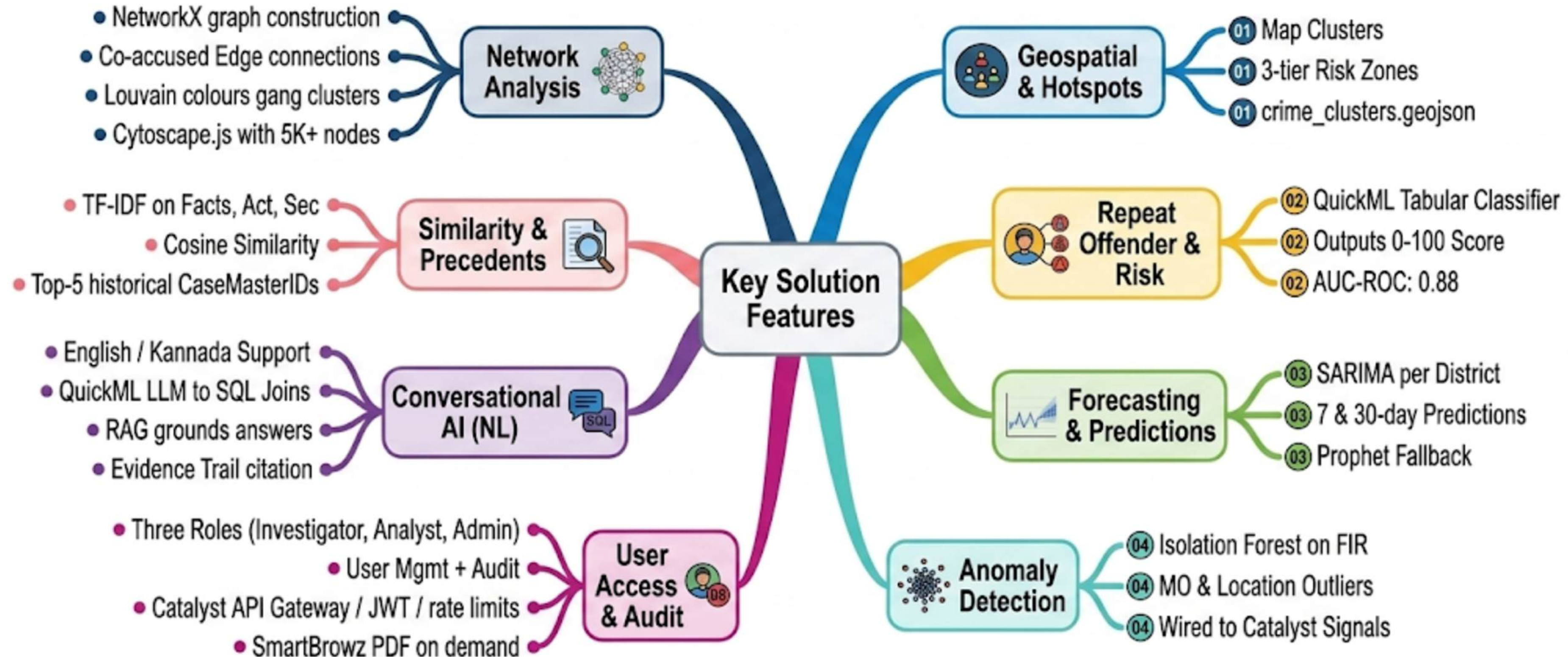
Kannada Voice I/O — Field officers speak Kannada into the mic. Zia STT transcribes, Zia Translation converts for the LLM, and Zia TTS reads the answer back in Kannada. No typing required.



Real-Time Anomaly Alerts — When Zia AutoML detects a crime spike, a Catalyst Signal fires instantly: Push Notification to the dashboard UI + Mail alert to the responsible district investigator. No page refresh, no delay.



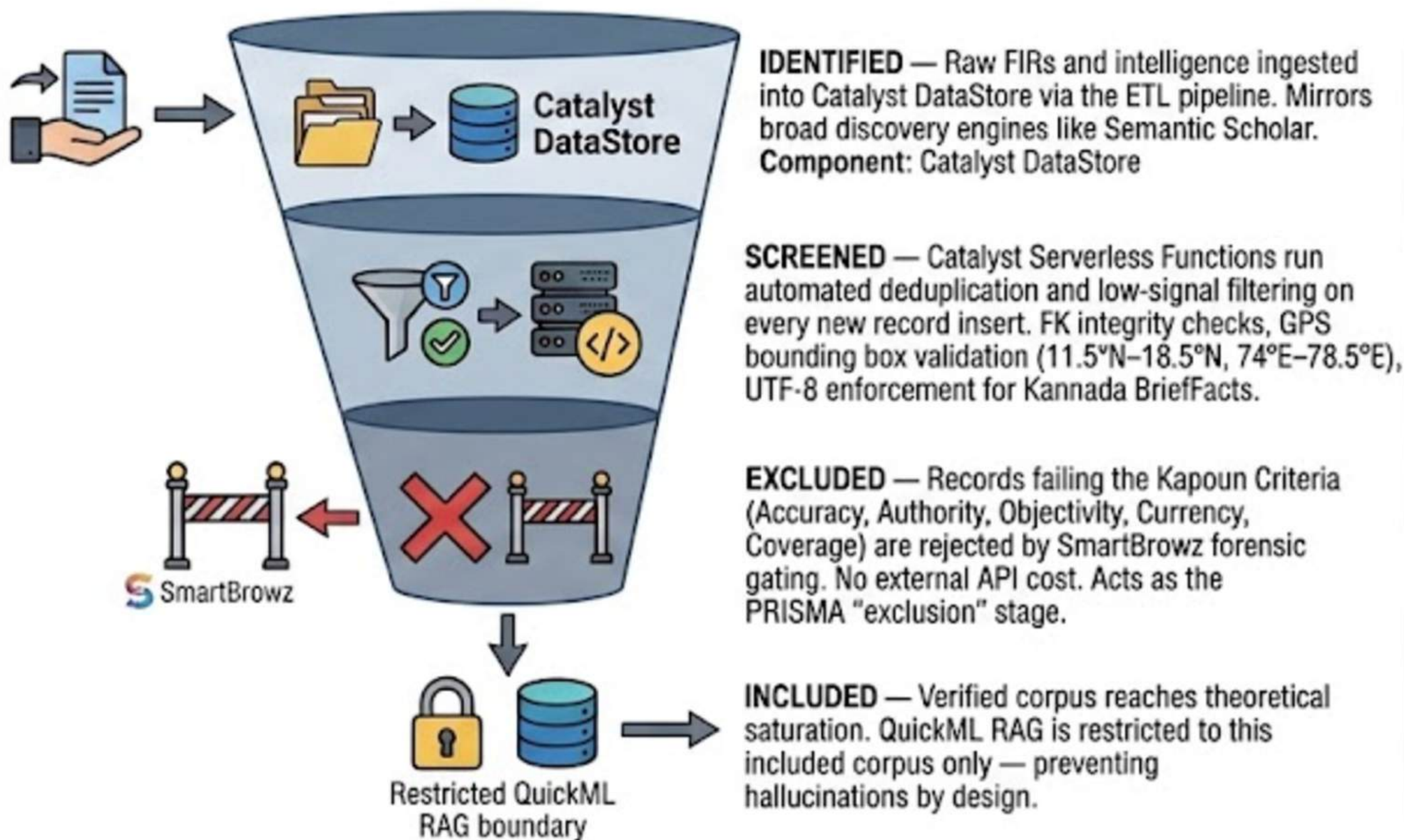
List of features offered by the solution





Process flow diagram or Use-case diagram

The PRISMA Funnel Process Flow



Key User Roles and Use-Cases



Role: Investigator

Day-to-day FIR lookup and query. Can chat, view the crime map, drill into case details, and export Evidence Trail PDFs.



Role: Analyst

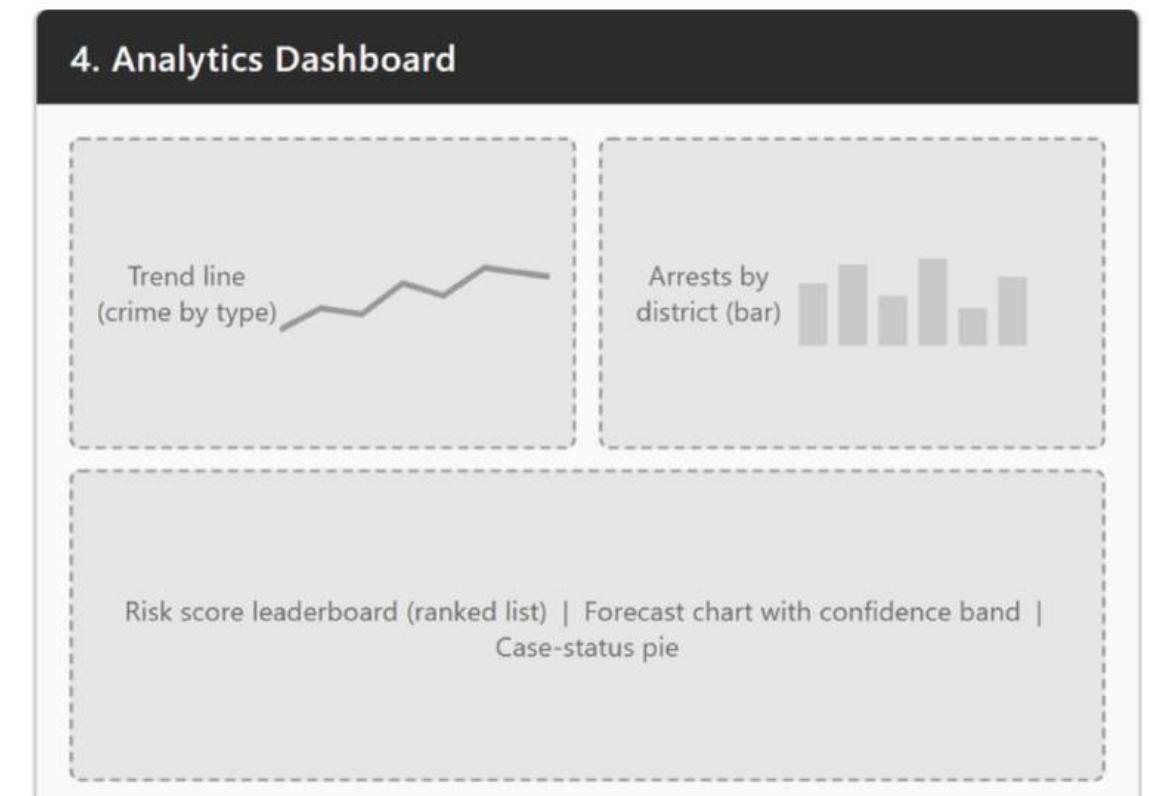
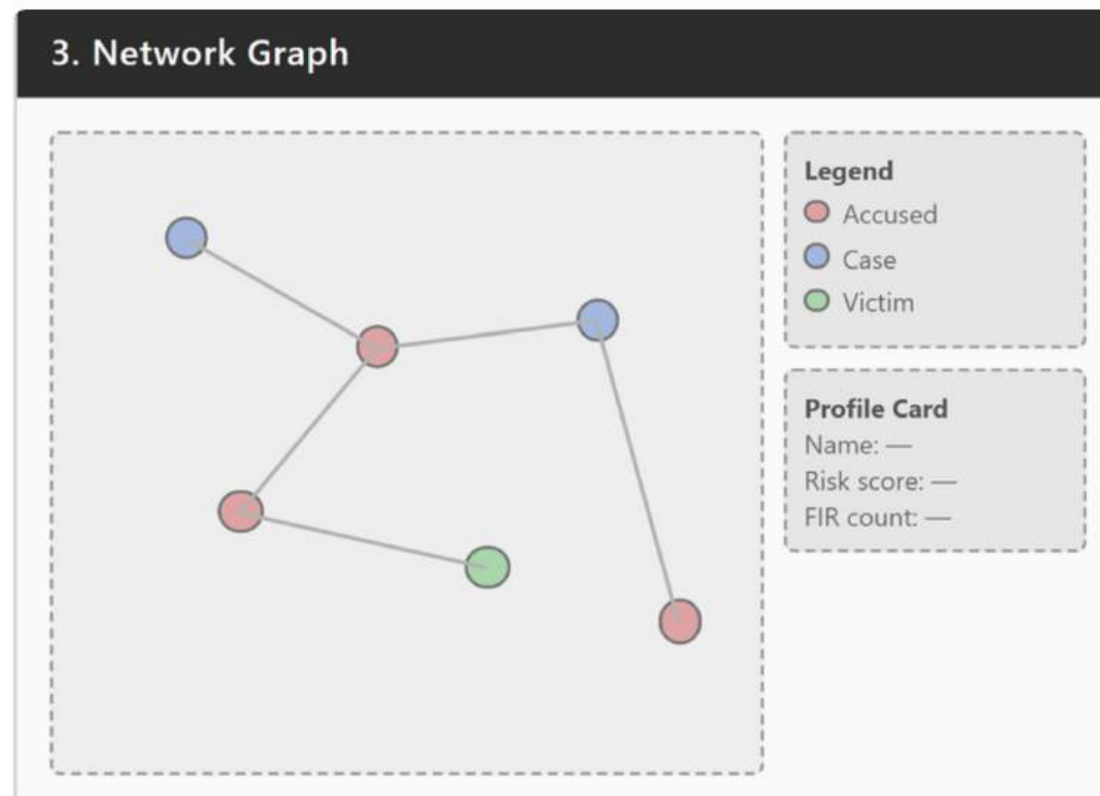
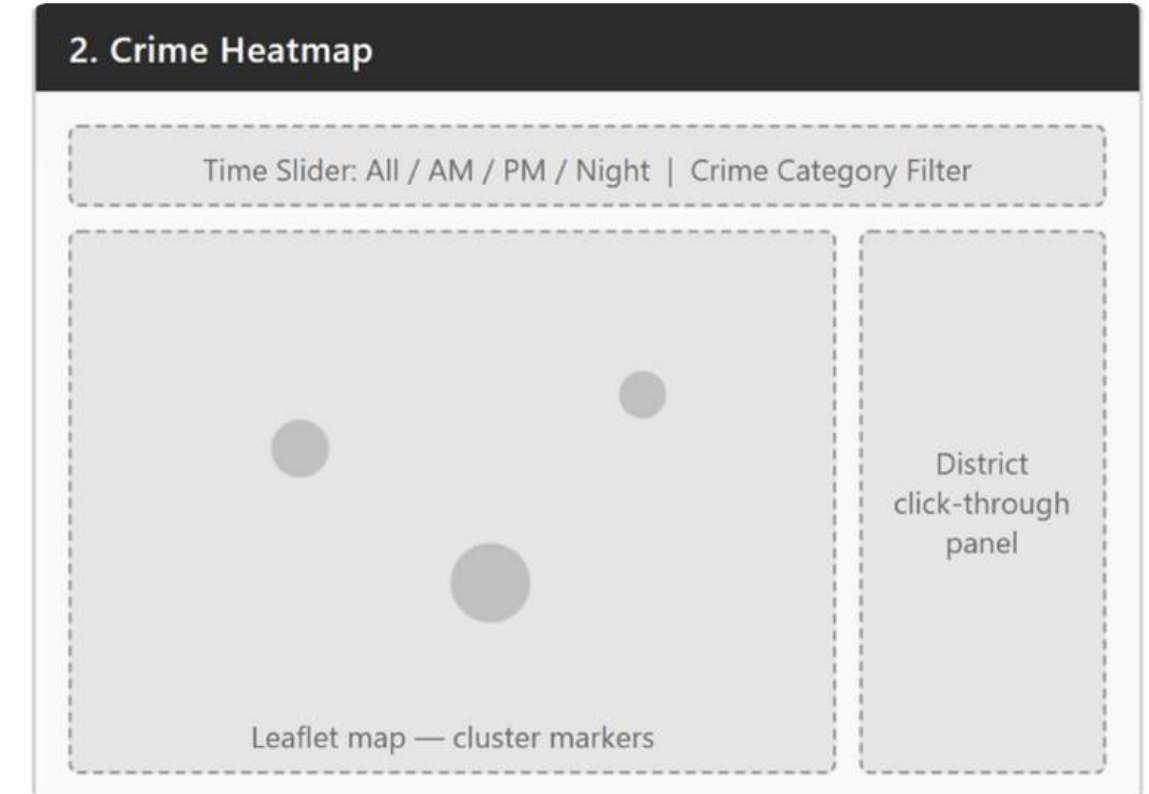
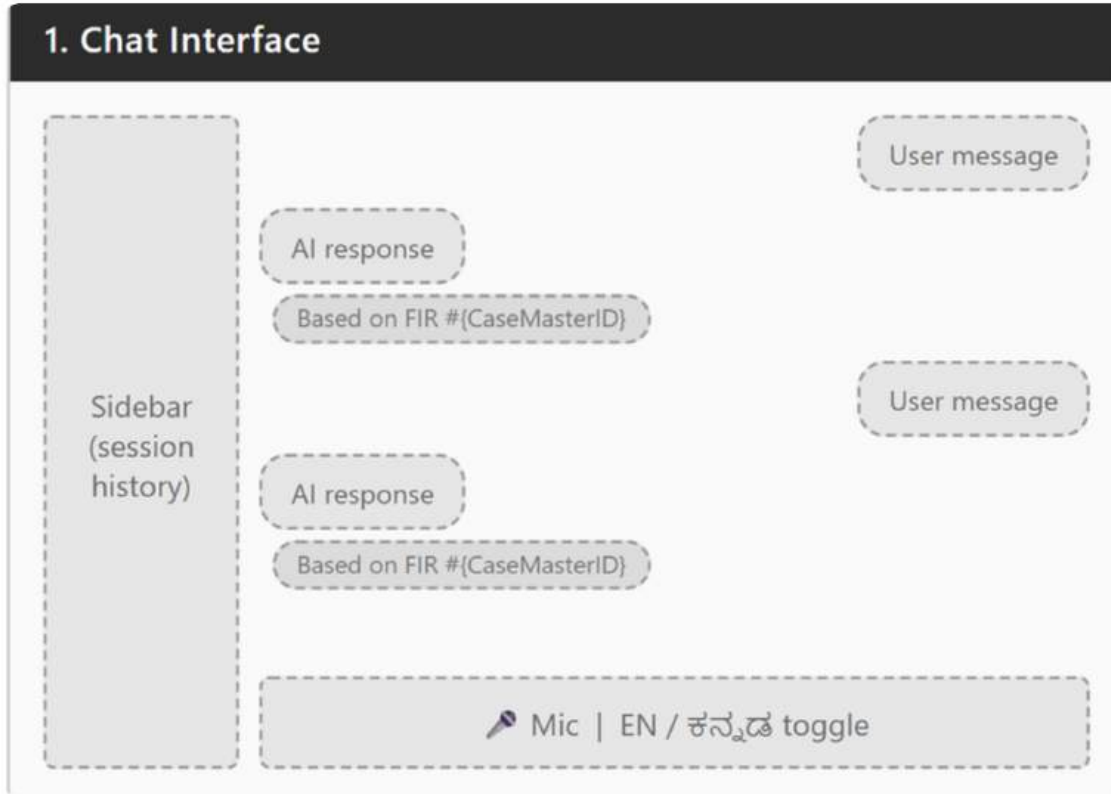
Strategic analytics view. Access to forecasting charts, district rollups, criminal network graph, and risk score leaderboard.



Role: Admin

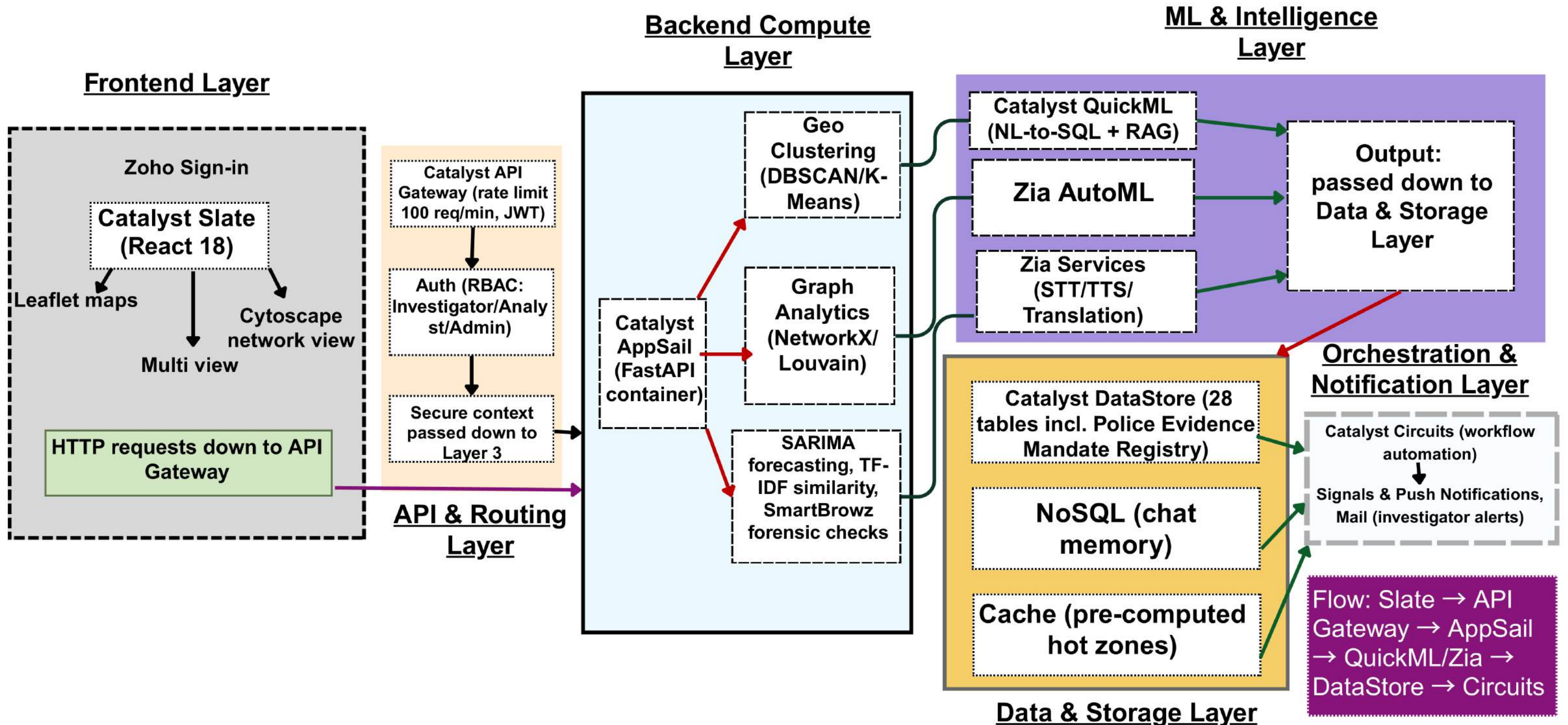
Full system access. Manages user accounts, reviews the complete audit log, and can unmask victim PII fields masked for lower roles.

Wireframes/Mock diagrams of the proposed solution (optional)





Architecture diagram of the proposed solution





Technologies to be used in the solution

Frontend



GPS Points Render



10K+
in under 2s.



Network Nodes

5K+
Smooth

Backend



API Container



FastAPI / Docker
multi-stage



p95 Latency



< 500ms
Target

ML / Analytics



Cloud GPU



None Required
(AppSail Compute)

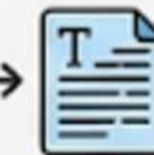
Core Python Models



SARIMA



Louvain



TF-IDF

Platform



aws

100%
Sovereign

Sovereign Services



**All Catalyst
Native**

List down Catalyst Services being used in the solution

Compute

- AppSail, Functions
- Used for:– **FastAPI backend; data validation**

Intelligence

- QuickML, Zia Services
- Used for:– **LLM + RAG chat; STT /TTS/ Translation; risk & anomaly scoring**

Storage

- DataStore, NoSQL, Stratus, Cache
- Used for:– **28 relational tables; chat sessions; district-stat caching**



Experience

- Slate, Authentication, API Gateway
- Used for:– **React hosting; RBAC login; rate-limit + JWT**

Automation

- Circuits, Signals, Push Notifications
- Used for:– **Query orchestration; anomaly event routing; scheduled cleanup; alert emails**

Reports

- SmartBrowz
- Used for:– **Evidence verification (Kapoun criteria) + PDF export**

Estimated implementation cost (optional)

Hackathon build:

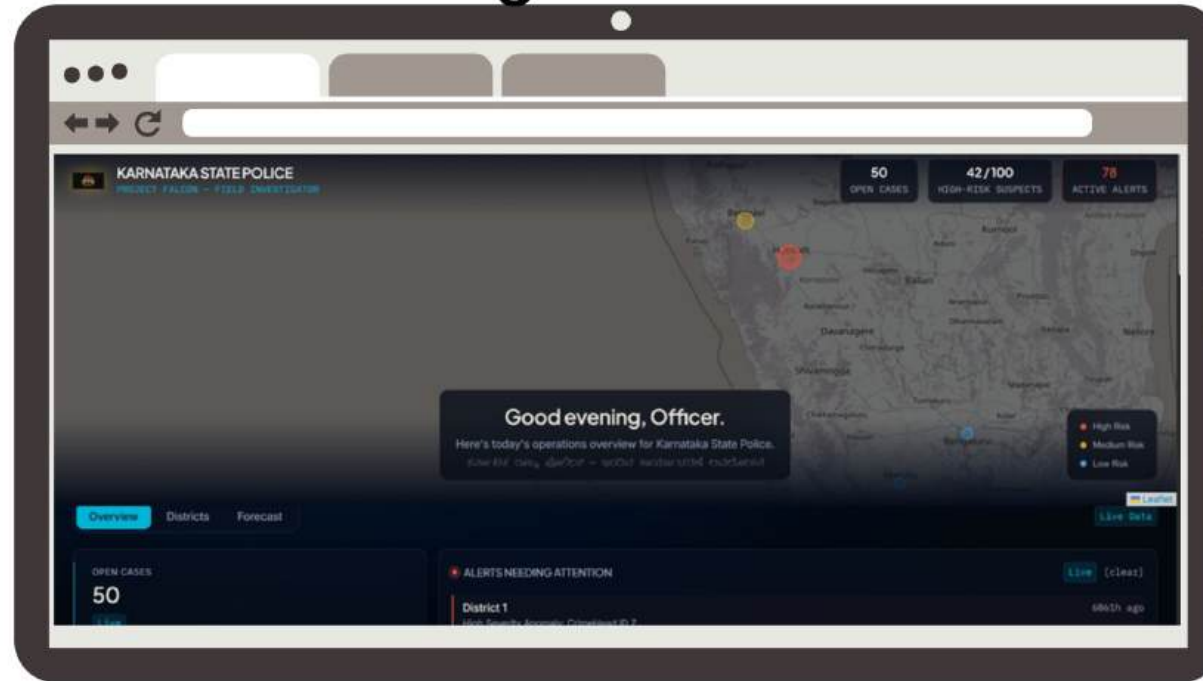
Run entirely on Zoho Catalyst credits (free tier)	— ₹0 additional infra cost during the Hackathon level project set up.
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Production build:

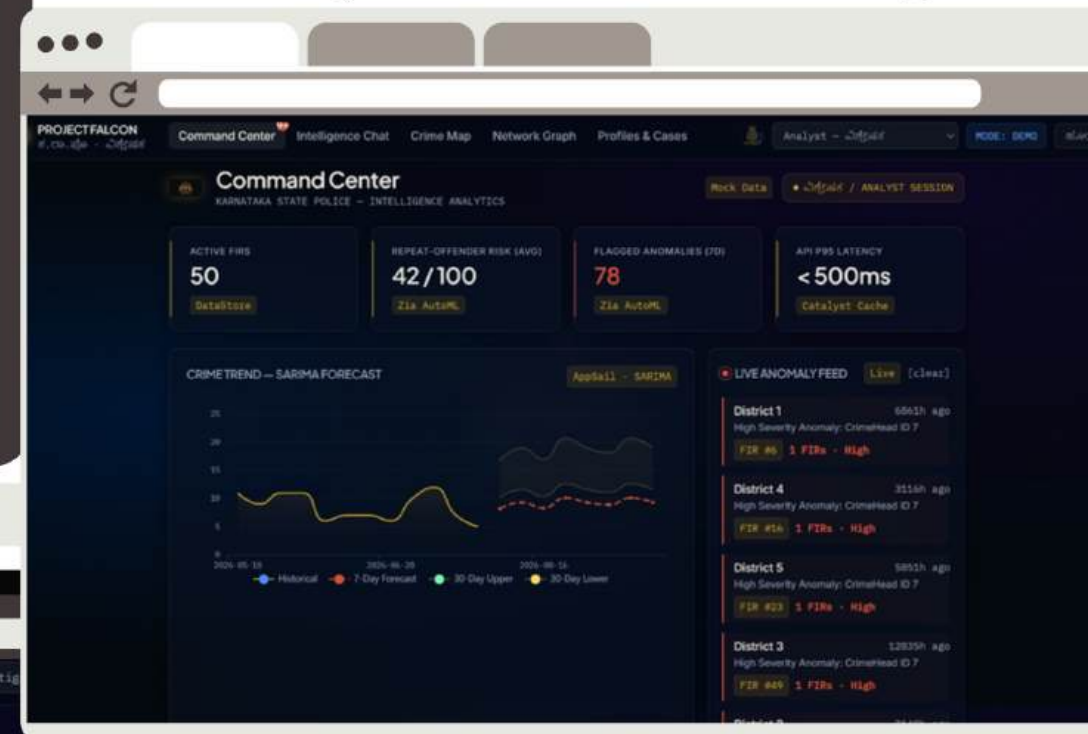
Zoho Catalyst Service	Usage in Project Falcon	Estimated Monthly Usage	Estimated Cost/Month
Catalyst Subscription	Production hosting and services	Premium production tier	\$50
AppSail Compute	Backend APIs and ML processing	Backend APIs and ML processing	\$200
DataStore	FIR, case, and investigation records	~200 GB storage	\$50
QuickML	Crime prediction and offender risk scoring	~100K inference calls	\$100
Zia Services	Kannada-English voice assistant	~20K voice requests	\$50
Cache + Stratus	Dashboard caching and evidence files	~500 GB storage	\$50
Total Estimated Cost	State-wide scalable deployment estimate	Optimized production usage	≈ \$500/month

Snapshots of the prototype

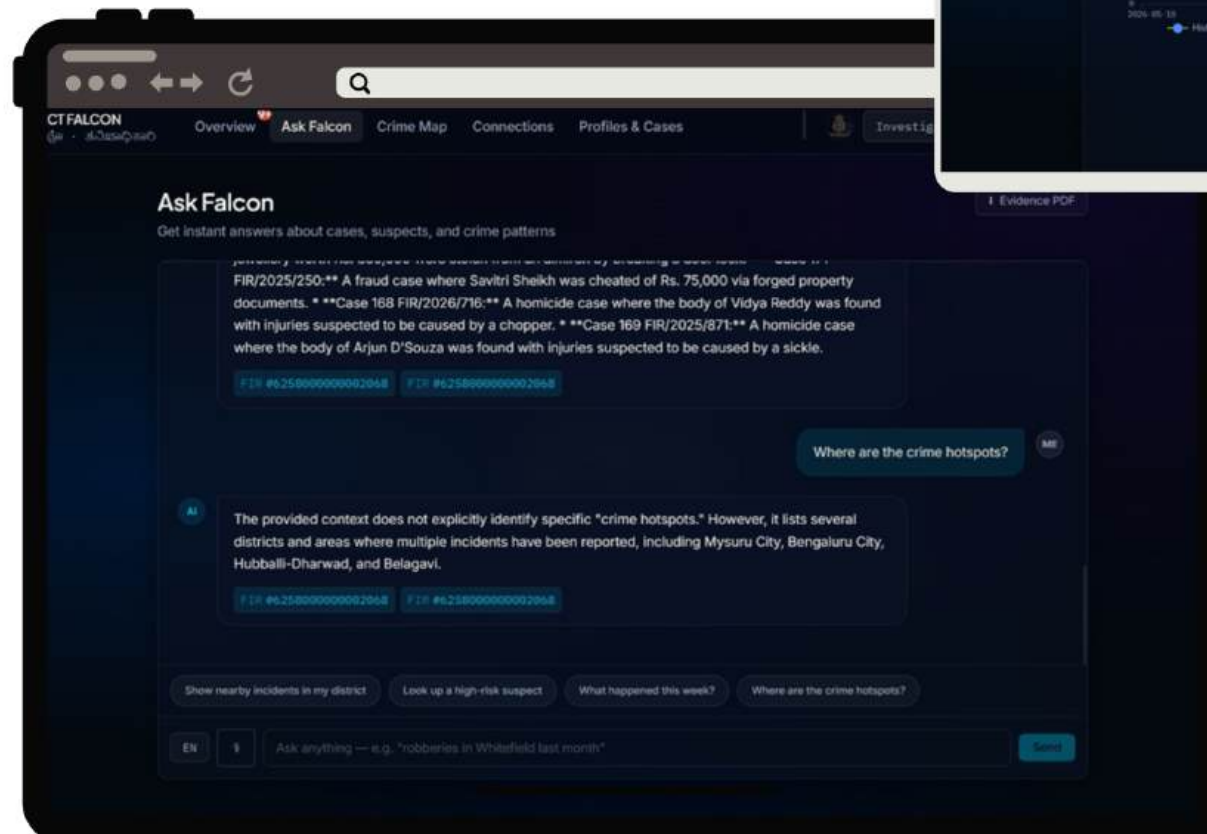
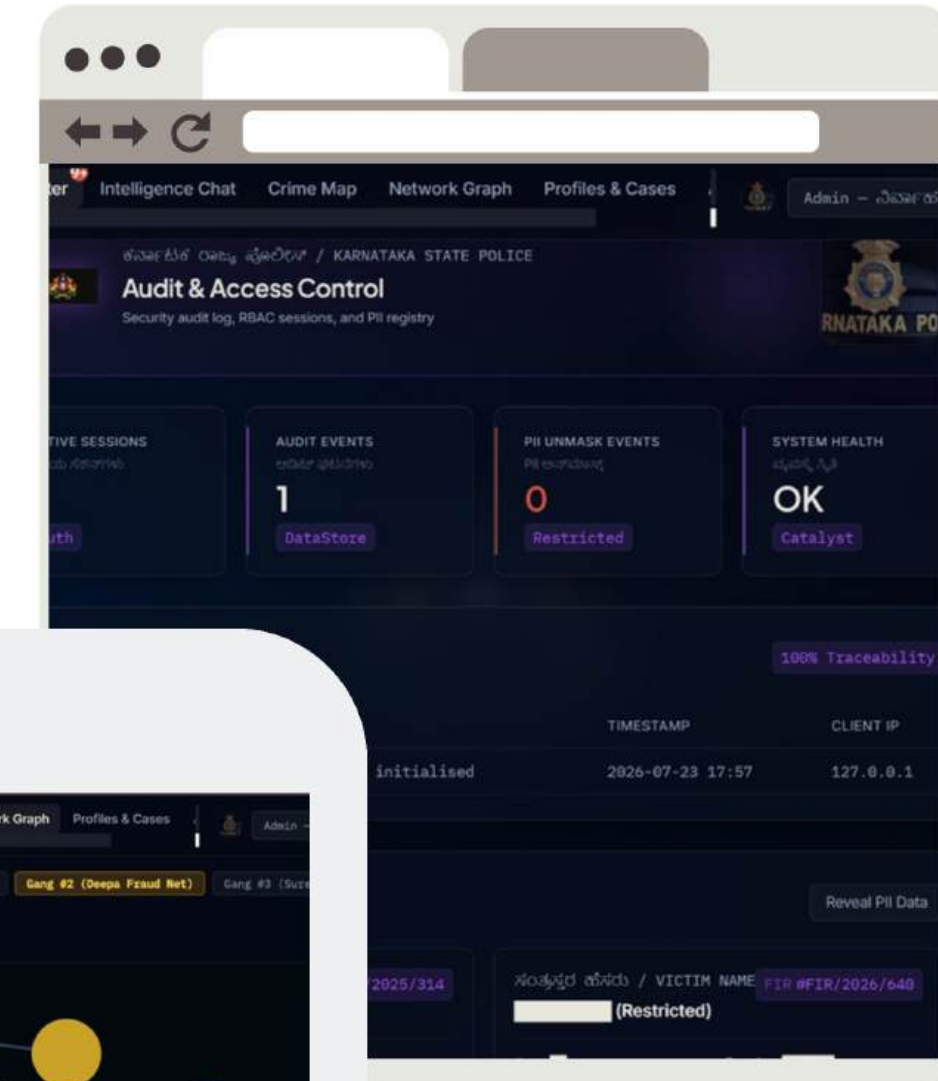
Investigator Dashboard



Analytics and forecasting

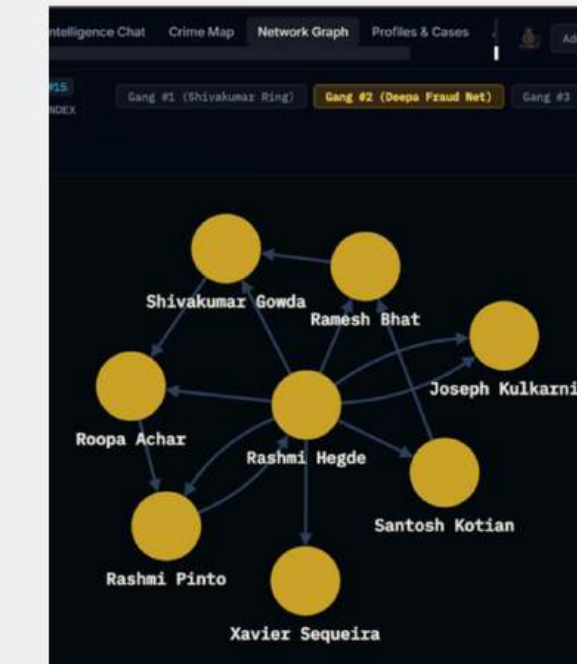


Audit and Security



Falcon AI ChatBot

Crime network graph



Prototype Performance report/Benchmarking

METRICS	TARGET	ACHIEVED
API P95 latency	< 500 ms	Served via Catalyst Cache
Geospatial query (100k points)	< 1.2 s	PostgreSQL indexing on DataStore
NL-to-SQL translation	< 2.5 s	QuickML response time
RAG document retrieval	< 1.5 s	QuickML fetch and response time
Uptime target	99.90%	Native Catalyst SLA
Repeat-offender risk model AUC-ROC	> 0.75	0.88 achieved
Map render (10K GPS points)	< 2 s	Leaflet + AppSail
Network graph (5K nodes)	No browser freeze	Cytoscape.js
Concurrent LLM queries	50 without crash	AppSail + Circuits
Cache warm-hit latency	< 100 ms	~5–10 ms observed



Provide links to your:

- **GitHub Public Repository** : <https://github.com/MUKUL-PRASAD-SIGH/Project-Falcon.git>
- **Demo Video Link (3 Minutes)**: <https://www.youtube.com/watch?v=F9OGUbJrP9w>
- **Deployed Link**: <https://auth-60079106947.development.catalystserverless.in/app/index.html>



Future Development



SOCIOLOGICAL & SOCIO-ECONOMIC OVERLAY

- Extend partial implementation to full correlation dashboard.
- Correlate urbanisation & sociological data.

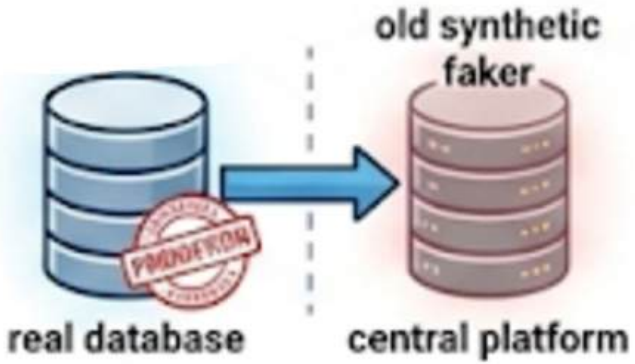


VS



REAL FINANCIAL ANALYSIS

- Needs real transaction/financial data source.
- Replace synthetic schema with live financial feeds.



REAL KSP DATASET INGESTION

- Ingest real production dataset.
- Move beyond current synthetic (Faker-generated) FIRs.



EXPAND KANNADA NLP & VOICE

- Deepen Kannada NLP/voice accuracy.
- Extend RAG saturation across the full corpus.



EXTEND SIGNALS TO SMS

- Add SMS channel alerting for field officers.
- Ensure access without constant dashboard check.



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THANK YOU